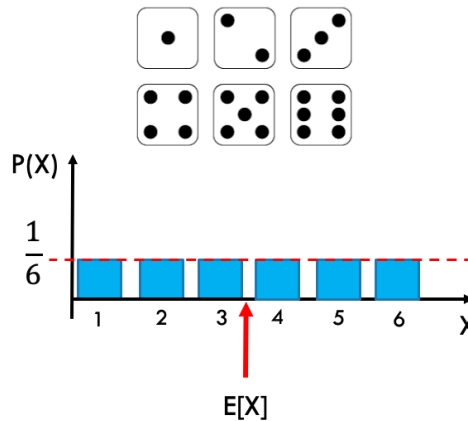


Reading 6.1: Graduates Case Study

Before we analyse the case, we need a refresher on the concept of the Expected Value of a random variable X : $E[X]$. To understand this, imagine rolling a die. Each face has an equal chance of appearing with probability $1/6$.



The expected value of a random variable X is defined as the weighted sum of all possible outcomes of an experiment. The weights are the probabilities of occurrence for each of the outcomes. Applied to our 'roll a die' experiment the expected value is: $\frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \frac{1}{6} \cdot 3 + \frac{1}{6} \cdot 4 + \frac{1}{6} \cdot 5 + \frac{1}{6} \cdot 6 = 3.5$

Let us look at a scenario where a high school graduate is tasked to decide about going to college or not. Assume following general statistics for high school graduates:

1. 40% start college immediately after high school graduation
2. 40% take a 1-year break to travel the world, and start college immediately thereafter
3. 20% skip college and start working immediately after high school graduation

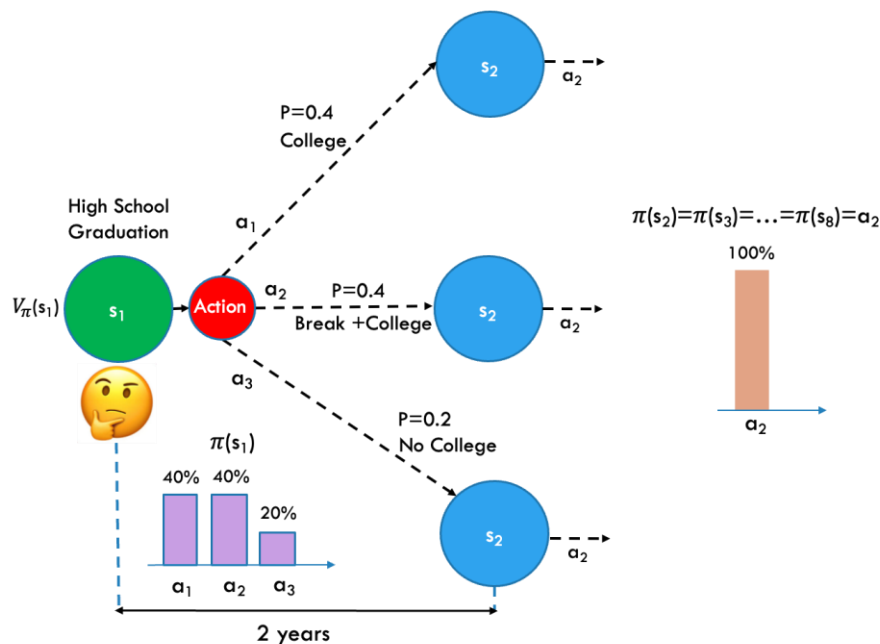
Assume that we have 8 states (s_1 to s_8) and that all states are 2 years apart. The state s_1 reflects high school graduation. Our policy $\pi(s_1)$ is stochastic with probabilities (40,40,20) and from s_2 onwards we move to the right in 2-year blocks, under a deterministic policy. We assume that college is 4 years.

We further assume that:

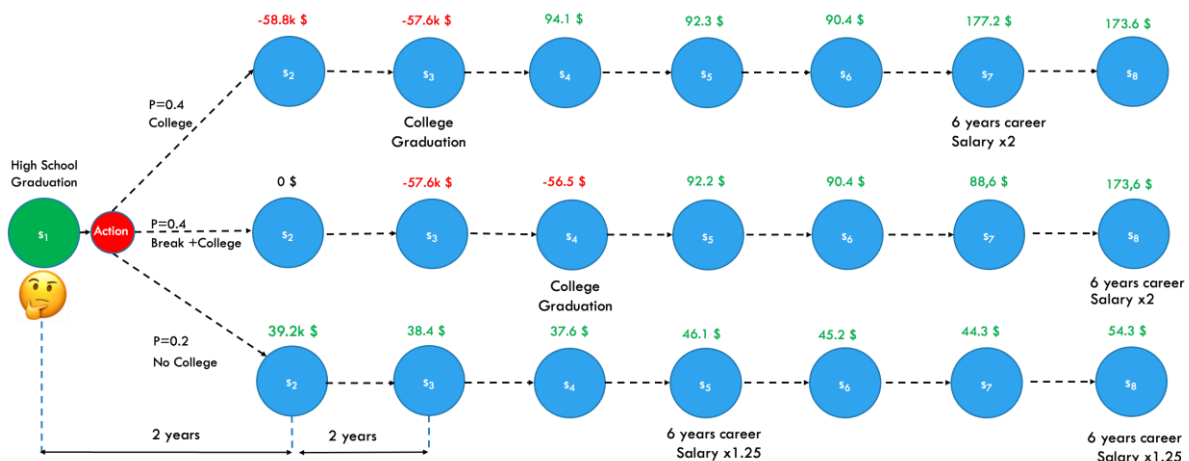
- college tuition fees are \$120k spread over 4 years
- the starting salary of a college graduate is \$50k per annum, doubling every 6 years
- the starting salary of a high school graduate is \$20k per annum, increasing every 6 years with 25%

When observing the different states, the reward represents the cash in- or outflows that the graduate receives upon arrival in a state. The graduate student becoming a college student immediately after

graduation will see a cash outflow of -\$60k when arriving in s_2 . Graduates skipping college earn \$40k in the first 2-year block.



We note that, for graduates entering the job market immediately after graduation, the reward in state s_2 is only \$39.2k even though the graduate earns \$20k per annum. The reason is that cash received now is worth more than receiving the same amount 2 years from now. To account for the time value of cash flows we introduce a discount factor γ and we set $\gamma = 0.98$ for each 2-year block. Applying the discount factor to our 'no college' scenario results in a cash inflow of $0.98 \times \$40k = \$39.2k$ in s_2 .



The cumulative cash flow in a state is the sum of all cash flows received up to and including that state. Looking at cumulative cash flows, we see that, in state s_8 , that the high school graduate, starting college immediately upon graduation, received \$511.1k. It confirms that short term choices have long term consequences. In s_7 , 12 years after graduation, the College scenario beats the No College scenario, and the gap will only increase from there onwards. The Break + College scenario surpasses the No College scenario in s_8 .

	State 2	State 3	State 4	State 5	State 6	State 7	State 8
College	-58.8	-116.4	-22.3	69.9	160.3	337.5	511.1
Break + College	0.0	-57.6	-114.1	-21.9	68.5	157.1	330.7
No College	39.2	77.6	115.3	161.4	206.6	250.9	305.1

The discounting of the cash flows method allows us to express all future cash flows in today's terms. For example, the cumulative discounted cash flow in the state s4 for the No College scenario is: (\$39.2+\$38.4+\$37.6) or \$115.3k. The College/No College example explains why, in RL, we are mainly interested in the long-term potential of a state.

We started this exercise because we want to find a way to express the potential of the state s1 given a policy π . First, let's define the return of the state s1 as $G(s_1)$ as the sum of the current and future discounted rewards:

$$G(s_t) = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots$$

↑ immediate reward
 ⏟ discounted future rewards

In our example, the return is the total cash collected by the agent arriving in state s8 in each of the scenarios:

- $G(s_1)$ in the College Scenario: 511.1k
- $G(s_1)$ in the Break + College Scenario: 330.7k
- $G(s_1)$ in No College Scenario: 305.1k

The value of the state s1 is defined as the expected return for an agent starting its journey in s1 and following the policy π from there onwards. Mathematically this is expressed by following formula.

$$V_{\pi}(s_1) = E_{\pi}[G(s_1)]$$

↓ State Value
 ↓ Return

↑ Expected Value

$$V(s_1) = 40\% * (\text{return of the College scenario}) + 40\% * (\text{return of the Break + College scenario}) + 20\% * (\text{return of the No College scenario}) = 204.4 + 132.3 + 61.0 = 397.7k$$

The intuition behind $V(s_1)$ is that a high school graduate, on average, and before making a college/no college decision, can expect to earn \$397.7k in cash in the 14-year period after high school graduation, expressed in today's terms.